



Quality Control Report

Report ID: QC-S89921-01
Generated on 07/31/2026

Report to

Attention: Cindy Tomaszewski
City of Cadillac - WWTP
1121 Plett Rd
Cadillac, MI 49601

Phone: 231-775-2368 FAX:

Report Produced by

Merit Laboratories
2680 East Lansing Drive
East Lansing, MI 48823

Phone: (517) 332-0167 FAX: (517) 332-6333

Report Summary

Lab Sample ID(s): S89921.01
Project: Monitoring
Submitted Date/Time: 07/23/2026 10:20
Sampled by: Taysha Laczko
P.O. #:

QC Report Sections

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Report Flag Descriptions

*: QC result is outside of indicated control limits
W: Surrogate result not applicable due to sample dilution

I certify that this data package is in compliance with the terms and conditions of the program, and project, and contractual requirements both technically and for completeness. Release of the data contained in this hardcopy data package and its computer-readable data submitted has been authorized by the Quality Assurance Manager and his/her designee, as verified by the following signature.

Barbara Ball
Quality Assurance Manager

QC Report - Analysis Summary

Lab Sample ID: S89921.01

Sample Tag: Digester #4

Collected Date/Time: 07/22/2026 13:30

Matrix: Sludge

COC Reference: 189611

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
<i>Inorganics</i>						
Total Solids	SM2540B	07/23/26 18:39	TS260723A	TS260723A	No	BLK/LCS/DUP
<i>Organics - Volatiles</i>						
PFAs EPA Method 1633	EPA 1633A	07/28/26 18:59	CI260728_1633S	E1633260727S1	Yes	BLK/LCS
PFAs EPA Method 1633 (Replicate 01)	EPA 1633A	07/28/26 18:38	CI260728_1633S	E1633260727S1	Yes	BLK/LCS

QC Report - Prep Batch Summary

Inorganics, Prep Batch ID: TS260723A

Surrogates: No, QC Types: BLK/LCS/DUP

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S89921.01	Total Solids	SM2540B	07/23/26 18:39	TS260723A

Organics - Volatiles, Prep Batch ID: E1633260727S1

Surrogates: Yes, QC Types: BLK/LCS

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S89921.01	PFA's EPA Method 1633	EPA 1633A	07/28/26 18:59	CI260728_1633S
S89921.01	PFA's EPA Method 1633 (Replicate 01)	EPA 1633A	07/28/26 18:38	CI260728_1633S

QC Report - Surrogates per Lab Sample

Lab Sample ID: S89921.01

Sample Tag: Digester #4

Collected Date/Time: 07/22/2026 13:30

Matrix: Sludge

COC Reference: 189611

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: Cl260728_1633S, Run Date: 07/28/2026 18:59, Matrix: SO, Dilution: 17

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		107.4	5	130
M5PFPeA-S		112.6	35	130
M2-4:2FTSA-S		213.8	40	300
M5PFHxA-S		112.7	40	130
M3PFBS-S		104.9	40	150
MHFPO-DA-S		85.5	40	130
M4PFHpA-S		110.1	40	130
M2-6:2FTSA-S		236.3	40	300
M8PFOA-S		106.8	40	130
M3PFHxS-S		108.1	40	140
M9-PFNA-S		107.1	40	145
M2-8:2FTSA-S		234.8	40	300
d3N-MeFOSAA-S		129.0	30	150
M6PFDA-S		107.6	40	130
d5EtFOSAA-S	*	152.9	20	140
M8PFOS-S		101.5	40	130
M7PFUnDA-S		58.4	40	130
MPFDoDA-S	*	29.9	40	130
M8FOSA-S		118.5	20	140
M2PFTeDA-S		12.6	10	160
d7-N-MeFOSE-M-S		47.0	25	130
d-N-MeFOSA-M-S		34.3	20	130
d9-N-EtFOSE-M-S		43.5	20	130
d-N-EtFOSA-M-S		28.4	20	130
M6PFTTrDA-S	*	18.3	20	150

Organics - Volatiles, Analysis: PFAs EPA Method 1633 (Replicate 01)

Run in Batch: Cl260728_1633S, Run Date: 07/28/2026 18:38, Matrix: SO, Dilution: 120

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		109.9	5	130
M5PFPeA-S		110.3	35	130
M2-4:2FTSA-S		138.9	40	300
M5PFHxA-S		110.7	40	130
M3PFBS-S		108.0	40	150
MHFPO-DA-S		102.5	40	130
M4PFHpA-S		106.5	40	130
M2-6:2FTSA-S		127.7	40	300
M8PFOA-S		103.9	40	130
M3PFHxS-S		108.7	40	140
M9-PFNA-S		106.0	40	145
M2-8:2FTSA-S		121.1	40	300
d3N-MeFOSAA-S		119.9	30	150
M6PFDA-S		109.2	40	130
d5EtFOSAA-S		111.2	20	140
M8PFOS-S		105.2	40	130
M7PFUnDA-S		91.8	40	130
MPFDoDA-S		74.1	40	130

QC Report - Surrogates per Lab Sample

M8FOSA-S	104.4	20	140
M2PFTeDA-S	56.0	10	160
d7-N-MeFOSE-M-S	101.8	25	130
d-N-MeFOSA-M-S	96.1	20	130
d9-N-EtFOSE-M-S	95.5	20	130
d-N-EtFOSA-M-S	82.2	20	130
M6PFTrDA-S	75.7	20	150

QC Report - Surrogates per QC Sample

Organics - Volatiles, Prep Batch ID: E1633260727S1

QC Types: BLK/LCS

Blank (BLK)

Lab Sample ID: CI260728_1633S.BLK260727

Run in Batch: CI260728_1633S, Run Date: 07/28/2026 17:13, Prep Date: 07/27/2026, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		106.1	5	130
M5PFPeA-S		111.5	35	130
M2-4:2FTSA-S		111.4	40	300
M5PFHxA-S		107.5	40	130
M3PFBS-S		105.5	40	150
MHFPO-DA-S		109.0	40	130
M4PFHpA-S		105.6	40	130
M2-6:2FTSA-S		106.4	40	300
M8PFOA-S		104.4	40	130
M3PFHxS-S		105.1	40	140
M9-PFNA-S		102.1	40	145
M2-8:2FTSA-S		92.4	40	300
d3N-MeFOSAA-S		87.2	30	150
M6PFDA-S		105.4	40	130
d5EtFOSAA-S		78.8	20	140
M8PFOS-S		100.3	40	130
M7PFUnDA-S		97.4	40	130
MPFDoDA-S		86.9	40	130
M8FOSA-S		89.9	20	140
M2PFTeDA-S		73.0	10	160
d7-N-MeFOSE-M-S		75.4	25	130
d-N-MeFOSA-M-S		69.3	20	130
d9-N-EtFOSE-M-S		69.8	20	130
d-N-EtFOSA-M-S		64.4	20	130
M6PFTTrDA-S		74.4	20	150

Laboratory Control Sample (LCS)

Lab Sample ID: CI260728_1633S.LCS260727

Run in Batch: CI260728_1633S, Run Date: 07/28/2026 17:35, Prep Date: 07/27/2026, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		113.7	5	130
M5PFPeA-S		118.3	35	130
M2-4:2FTSA-S		116.7	40	300
M5PFHxA-S		114.1	40	130
M3PFBS-S		115.1	40	150
MHFPO-DA-S		112.9	40	130
M4PFHpA-S		112.2	40	130
M2-6:2FTSA-S		108.9	40	300
M8PFOA-S		112.9	40	130
M3PFHxS-S		114.6	40	140
M9-PFNA-S		111.7	40	145
M2-8:2FTSA-S		102.8	40	300
d3N-MeFOSAA-S		98.4	30	150
M6PFDA-S		103.7	40	130
d5EtFOSAA-S		90.8	20	140
M8PFOS-S		112.0	40	130
M7PFUnDA-S		101.4	40	130
MPFDoDA-S		90.7	40	130
M8FOSA-S		101.4	20	140

QC Report - Surrogates per QC Sample

M2PFTeDA-S	81.5	10	160
d7-N-MeFOSE-M-S	86.9	25	130
d-N-MeFOSA-M-S	85.2	20	130
d9-N-EtFOSE-M-S	81.5	20	130
d-N-EtFOSA-M-S	78.2	20	130
M6PFTrDA-S	85.5	20	150

QC Report - Surrogates per QC Sample

Laboratory Control Sample (LCS)

Lab Sample ID: CI260728_1633S.LLLCS26072

Run in Batch: CI260728_1633S, Run Date: 07/28/2026 17:56, Prep Date: 07/27/2026, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		102.8	5	130
M5PFPeA-S		103.5	35	130
M2-4:2FTSA-S		108.4	40	300
M5PFHxA-S		103.2	40	130
M3PFBS-S		102.4	40	150
MHFPO-DA-S		106.9	40	130
M4PFHpA-S		101.3	40	130
M2-6:2FTSA-S		99.4	40	300
M8PFOA-S		102.5	40	130
M3PFHxS-S		99.6	40	140
M9-PFNA-S		97.4	40	145
M2-8:2FTSA-S		94.7	40	300
d3N-MeFOSAA-S		98.5	30	150
M6PFDA-S		98.7	40	130
d5EtFOSAA-S		92.7	20	140
M8PFOS-S		102.6	40	130
M7PFUnDA-S		95.5	40	130
MPFDoDA-S		88.1	40	130
M8FOSA-S		93.1	20	140
M2PFTeDA-S		76.0	10	160
d7-N-MeFOSE-M-S		79.2	25	130
d-N-MeFOSA-M-S		76.8	20	130
d9-N-EtFOSE-M-S		73.5	20	130
d-N-EtFOSA-M-S		68.1	20	130
M6PFTTrDA-S		79.7	20	150

QC Report - Internal Standards per Lab Sample

Lab Sample ID: S89921.01

Sample Tag: Digester #4

Collected Date/Time: 07/22/2026 13:30

Matrix: Sludge

COC Reference: 189611

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI260728_1633S, Run Date: 07/28/2026 18:59, Matrix: SO, Dilution: 17

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		93.1	50	200.0
M2PFHxA		94.9	50	200.0
M4PFOA		94.2	50	200.0
M5-PFNA		96.4	50	200.0
M4PFOS		88.0	50	200.0
M2PFDA		89.5	50	200.0
O2PFHxS		95.5	50	200.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633 (Replicate 01)

Run in Batch: CI260728_1633S, Run Date: 07/28/2026 18:38, Matrix: SO, Dilution: 120

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		97.9	50	200.0
M2PFHxA		94.7	50	200.0
M4PFOA		100.9	50	200.0
M5-PFNA		97.5	50	200.0
M4PFOS		103.1	50	200.0
M2PFDA		103.4	50	200.0
O2PFHxS		95.5	50	200.0

QC Report - Internal Standards per QC Sample

Organics - Volatiles, Prep Batch ID: E1633260727S1

QC Types: BLK/LCS

Blank (BLK)

Lab Sample ID: CI260728_1633S.BLK260727

Run in Batch: CI260728_1633S, Run Date: 07/28/2026 17:13, Prep Date: 07/27/2026, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		93.2	50	200.0
M2PFHxA		88.5	50	200.0
M4PFOA		91.4	50	200.0
M5-PFNA		90.4	50	200.0
M4PFOS		94.3	50	200.0
M2PFDA		89.8	50	200.0
O2PFHxS		92.4	50	200.0

Laboratory Control Sample (LCS)

Lab Sample ID: CI260728_1633S.LCS260727

Run in Batch: CI260728_1633S, Run Date: 07/28/2026 17:35, Prep Date: 07/27/2026, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		90.7	50	200.0
M2PFHxA		87.2	50	200.0
M4PFOA		87.8	50	200.0
M5-PFNA		88.4	50	200.0
M4PFOS		90.3	50	200.0
M2PFDA		94.1	50	200.0
O2PFHxS		89.4	50	200.0

Laboratory Control Sample (LCS)

Lab Sample ID: CI260728_1633S.LLLCS26072

Run in Batch: CI260728_1633S, Run Date: 07/28/2026 17:56, Prep Date: 07/27/2026, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		91.5	50	200.0
M2PFHxA		88.9	50	200.0
M4PFOA		88.8	50	200.0
M5-PFNA		90.0	50	200.0
M4PFOS		92.3	50	200.0
M2PFDA		94.0	50	200.0
O2PFHxS		91.3	50	200.0

QC Report - Batch QC Results

Inorganics, Prep Batch ID: TS260723A

Surrogates: No, QC Types: BLK/LCS/DUP

Blank (BLK)

Lab Sample ID: TS260723A.LRB1

Run in Batch: TS260723A, Run Date: 07/23/2026 18:39, Prep Date: 07/23/2026, Matrix: Liquid, Dilution: 1

Analyte	Flags	Conc	RDL	Units
Total Solids		ND	1	%

Laboratory Control Sample (LCS)

Lab Sample ID: TS260723A.LCS1

Run in Batch: TS260723A, Run Date: 07/23/2026 18:39, Prep Date: 07/23/2026, Matrix: Liquid, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
Total Solids		100	90	110

Duplicate (DUP)

Lab Sample ID: TS260723A.DP1, Parent Sample ID: S89917.01

Run in Batch: TS260723A, Run Date: 07/23/2026 18:39, Prep Date: 07/23/2026, Matrix: Soil, Dilution: 1

Analyte	Flags	RPD	RPD CL
Total Solids		0	5

Duplicate (DUP)

Lab Sample ID: TS260723A.DP2, Parent Sample ID: S89928.01

Run in Batch: TS260723A, Run Date: 07/23/2026 18:39, Prep Date: 07/23/2026, Matrix: Soil, Dilution: 1

Analyte	Flags	RPD	RPD CL
Total Solids		0	5

QC Report - Batch QC Results

Organics - Volatiles, Prep Batch ID: E1633260727S1

Surrogates: Yes, QC Types: BLK/LCS

Blank (BLK)

Lab Sample ID: CI260728_1633S.BLK260727

Run in Batch: CI260728_1633S, Run Date: 07/28/2026 17:13, Prep Date: 07/27/2026, Matrix: SO, Dilution: 1

Analyte	Flags	Conc	RDL	Units
PFBA		ND	100	ng/kg
PFMPA		ND	100	ng/kg
FPrPA (3:3 FTCA)		ND	200	ng/kg
PFPeA		ND	100	ng/kg
PFMBA		ND	100	ng/kg
4:2 FTSA		ND	100	ng/kg
NFDHA		ND	100	ng/kg
PFHxA		ND	100	ng/kg
PFBS		ND	100	ng/kg
HFPO-DA		ND	100	ng/kg
FPePA (5:3 FTCA)		ND	200	ng/kg
PFEESA		ND	100	ng/kg
PFHpA		ND	100	ng/kg
PFPeS		ND	100	ng/kg
ADONA		ND	100	ng/kg
6:2 FTSA		ND	100	ng/kg
PFOA		ND	100	ng/kg
PFHxS		ND	100	ng/kg
FHpPA (7:3 FTCA)		ND	200	ng/kg
PFNA		ND	100	ng/kg
8:2 FTSA		ND	100	ng/kg
PFHpS		ND	100	ng/kg
PFDA		ND	100	ng/kg
N-MeFOSAA		ND	100	ng/kg
EtFOSAA		ND	100	ng/kg
PFOS		ND	100	ng/kg
PFUnDA		ND	100	ng/kg
9CL-PF3ONS		ND	100	ng/kg
PFNS		ND	100	ng/kg
PFDODA		ND	100	ng/kg
PFDS		ND	100	ng/kg
PFTTrDA		ND	100	ng/kg
11CL-PF3OUdS		ND	100	ng/kg
FOSA		ND	100	ng/kg
PFTeDA		ND	100	ng/kg
PFDOS		ND	100	ng/kg
NMeFOSE		ND	100	ng/kg
NMeFOSAM		ND	100	ng/kg
NEtFOSE		ND	100	ng/kg
NEtFOSAM		ND	100	ng/kg

QC Report - Batch QC Results

Organics - Volatiles, Prep Batch ID: E1633260727S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS

Laboratory Control Sample (LCS)

Lab Sample ID: CI260728_1633S.LCS260727

Run in Batch: CI260728_1633S, Run Date: 07/28/2026 17:35, Prep Date: 07/27/2026, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
PFBA		103.1	70	140
PFMPA		102.2	30	140
FPrPA (3:3 FTCA)		76.5	45	130
PFPeA		101.3	60	150
PFMBA		101.4	60	150
4:2 FTSA		100.6	60	150
NFDHA		94.9	60	155
PFHxA		103.4	65	140
PFBS		102.4	65	145
HFPO-DA		109.8	70	145
FPePA (5:3 FTCA)		83.9	60	130
PFEESA		110.3	70	140
PFHpA		102.2	65	145
PFPeS		104.1	55	160
ADONA		102.9	70	160
6:2 FTSA		99.8	55	200
PFOA		99.5	70	150
PFHxS		100.2	60	150
FHpPA (7:3 FTCA)		87.4	60	150
PFNA		98.9	70	155
8:2 FTSA		95.1	70	150
PFHpS		105.8	65	155
PFDA		103.0	70	155
N-MeFOSAA		96.2	65	155
EtFOSAA		100.1	65	165
PFOS		96.3	65	160
PFUnDA		103.1	70	155
9CL-PF3ONS		101.1	70	150
PFNS		93.7	55	140
PFDoDA		105.6	70	150
PFDS		84.0	40	155
PFTTrDA		96.1	65	150
11CL-PF3OUdS		87.2	45	160
FOSA		98.2	70	140
PFTeDA		103.8	65	150
PFDOS		86.0	25	160
NMeFOSE		99.5	70	140
NMeFOSAM		102.2	70	155
NEtFOSE		98.8	65	165
NEtFOSAM		99.6	70	140

QC Report - Batch QC Results

Organics - Volatiles, Prep Batch ID: E1633260727S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS

Laboratory Control Sample (LCS)

Lab Sample ID: CI260728_1633S.LLLCS26072

Run in Batch: CI260728_1633S, Run Date: 07/28/2026 17:56, Prep Date: 07/27/2026, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
PFBA		95.4	70	140
PFMPA		94.9	30	140
FPrPA (3:3 FTCA)		93.0	45	130
PFPeA		100.3	60	150
PFMBA		93.0	60	150
4:2 FTSA		94.5	60	150
NFDHA		87.6	60	155
PFHxA		98.9	65	140
PFBS		96.4	65	145
HFPO-DA		89.0	70	145
FPePA (5:3 FTCA)		69.7	60	130
PFEESA		101.8	70	140
PFHpA		98.9	65	145
PFPeS		104.1	55	160
ADONA		93.3	70	160
6:2 FTSA		95.5	55	200
PFOA		99.1	70	150
PFHxS		104.7	60	150
FHpPA (7:3 FTCA)		82.5	60	150
PFNA		103.3	70	155
8:2 FTSA		99.6	70	150
PFHpS		93.8	65	155
PFDA		101.0	70	155
N-MeFOSAA		84.7	65	155
EtFOSAA		101.3	65	165
PFOS		98.5	65	160
PFUnDA		97.9	70	155
9CL-PF3ONS		92.4	70	150
PFNS		93.1	55	140
PFDODA		103.5	70	150
PFDS		84.4	40	155
PFTTrDA		99.4	65	150
11CL-PF3OUdS		80.5	45	160
FOSA		95.6	70	140
PFTeDA		98.6	65	150
PFDOS		84.9	25	160
NMeFOSE		99.0	70	140
NMeFOSAM		98.8	70	155
NEtFOSE		93.2	65	165
NEtFOSAM		99.3	70	140

REPORT TO

CHAIN OF CUSTODY RECORD

INVOICE TO

CONTACT NAME Cindy Tomaszewski			
COMPANY City of Cadillac - WWTP			
ADDRESS 1121 Plett Rd			
CITY Cadillac		STATE MI	ZIP CODE 49601
PHONE NO. 231-775-2368		CELL NO.	P.O. NO.
E-MAIL ADDRESS Lab@cadillac-mi.net		QUOTE NO.	

CONTACT NAME		Cindy Tomaszewski		☒ SAME	
COMPANY					
ADDRESS					
CITY				STATE	ZIP CODE
PHONE NO.			E-MAIL ADDRESS		

ANALYSIS (ATTACH LIST IF MORE SPACE IS REQUIRED)

PROJECT NO./NAME	SAMPLER(S) - PLEASE PRINT/SIGN NAME <i>Taysha Laczko T. Lagko</i>
TURNAROUND TIME REQUIRED ☐ 1 DAY ☐ 2 DAYS ☐ 3 DAYS ☒ STANDARD ☐ OTHER _____	
DELIVERABLES REQUIRED ☒ LEVEL II ☐ LEVEL III ☐ LEVEL IV ☐ EDD ☐ OTHER _____	

MATRIX	W=WATER	GW=GROUNDWATER	WW=WASTEWATER	S=SOIL	L=LIQUID	SD=SOLID
CODE:	SL=SLUDGE	DW=DRINKING WATER	O=OIL	WP=WPIE	A=AIR	WS=WASTE

Containers & Preservatives

[illegible][illegible]

RELINQUISHED BY:	<i>T. Laaska</i>	DATE	TIME
SIGNATURE/ORGANIZATION	<i>City of Cadillac</i>	7-22-26	1345
RECEIVED BY:		DATE	TIME
SIGNATURE/ORGANIZATION			

RELINQUISHED BY:		DATE	TIME
SIGNATURE/ORGANIZATION			
RECEIVED BY:		DATE	TIME
SIGNATURE/ORGANIZATION			

RELINQUISHED BY: SIGNATURE/ORGANIZATION		DATE 7/23/26		TIME 1020	
RECEIVED BY: SIGNATURE/ORGANIZATION		DATE 7/23/26		TIME 1020	
SEAL NO.	SEAL INTACT YES ☐ NO ☐	INITIALS	☒ ICE (SOLID)	TEMP. ON ARRIVAL	5.8
SEAL NO.	SEAL INTACT YES ☐ NO ☐	INITIALS		☐ BLUE ICE	
			☐ ICE (MELTED)	☐ NONE	